Invisible Farms: Evaluating Irrigation Detection for Smallholder Farmers in Zambia

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Abstract

We present the first empirical evaluation and extension of the small-scale irrigation detection baselines from Boser et al. (2025), which maps dry-season irrigation in Zambia from Sentinel-2 time-series imagery using Random Forest and Gradient Boosting classifiers. Because the original pipeline depends on gated Google Earth Engine access, we first contribute a fully open, STAC-based data infrastructure replacement using Microsoft Planetary Computer, yielding identical downstream features with no authentication barrier. We reproduce the core Random Forest baseline under a matched protocol, achieving P=0.47, R=0.24, F1=0.32, AUPRC=0.41 on our held-out test split vs. the papers ́ reported metrics (not available; the original is a proposal without published baselines), and document all implementation ambiguities encountered. We then extend the study along four axes motivated by the deployment context of low-resource Sub-Saharan Africa: (1) *lightweight models* — smaller tree-based and distilled alternatives with <10× the original footprint, achieving F1=0.48 (Logistic Regression) at >3,000× smaller model size and 12× faster inference; (2) *robustness* — performance under simulated cloud gaps and cross-year temporal shift; (3) *fairness* — systematic accuracy gaps between smallholder (<2 ha) and large (>10 ha) irrigated parcels across Zambia's provinces; and (4) *explainability* — SHAP-based feature attribution revealing the relative importance of NDVI, NDWI, and EVI across dry-season months. Our fairness analysis reveals smallholder recall of 0.23 vs. 0.30 for large farms — the model misses 77% of irrigated small parcels, highlighting a practical risk for policy applications targeting the communities most in need of irrigation support. All code and a one-command Kaggle-based reproduction scaffold are released at `https://anonymous.4open.science/r/smallholder-irrigation-dataset-D08C/`.

1 Introduction and Motivation

Smallholder farmers account for the majority of dry-season irrigation activity in Zambia and much of Sub-Saharan Africa, yet they remain systematically undercounted in national and global irrigation inventories (Siebert et al., 2010; Food and Agriculture Organization of the United Nations, 2020). Accurate, fine-grained irrigation maps are a prerequisite for evidence-based water governance, climate-adaptation planning, and targeted agricultural support — but producing them at continental scale from field surveys is prohibitively expensive. Remote sensing offers a scalable alternative: Sentinel-2 time-series provide 10-meter, freely available multispectral imagery over the full dry season, and machine learning classifiers can in principle distinguish irrigated from rain-fed agriculture from vegetation index trajectories (Drusch et al., 2012; Thenkabail et al., 2009).

Boser et al. (2025) demonstrate this approach for Zambia, combining a hand-labeled polygon dataset with Sentinel-2 NDVI/NDWI/EVI time series to train Random Forest and Gradient Boosting classifiers, and show that the resulting maps correctly identify smallholder irrigation parcels too small to appear in official datasets. Their work is methodologically sound and socially relevant, but its pipeline depends on Google Earth Engine (GEE) service-account access, making it difficult for practitioners in low-resource institutions to reproduce or build upon. Our first contribution removes this barrier.

Beyond implementation, we ask: *are the original models fit for purpose in the deployment contexts where irrigation maps are most needed?* Specifically, we evaluate (1) whether smaller, faster models maintain adequate accuracy for the annotation-scarce regime common in Sub-Saharan Africa; (2) whether performance degrades under realistic data-quality shifts (cloud cover, cross-year); (3) whether model accuracy is equitable across farm sizes and provinces; and (4) whether model decisions can be explained in terms intelligible to local agronomists and NGO practitioners.

These extensions are not decorative: the original paper's primary motivation is informing climate-adaptation policy for smallholder communities (Lobell et al., 2011; Thornton & Lipper, 2014). A model that is systematically less accurate for small farms, or that a local practitioner cannot interrogate, is less useful for that purpose regardless of its aggregate F1.

Contributions.

1. A fully open, GEE-free data pipeline using Microsoft Planetary Computer STAC and SCL cloud masking, enabling one-command reproduction without authentication credentials (§4).
2. A documented reproduction of the Random Forest and Gradient Boosting baselines, with all hyperparameters, training times, and deviations from the original (§5).
3. Lightweight model extensions: smaller tree ensembles, depth-limited forests, and efficiency benchmarks (§6.1).
4. Robustness analysis under cloud-gap simulation and cross-year shift (§6.2).
5. Cross-region and smallholder-vs.-large fairness evaluation (§6.3).
6. SHAP-based explainability revealing feature attribution by month and parcel size (§6.4).

2 Related Work

Remote sensing for irrigation mapping. Vegetation index time series from Landsat and Sentinel-2 have been used to map irrigated area at regional and global scales (Thenkabail et al., 2009; Siebert et al., 2010). Most large-scale products target field sizes >5 ha; smallholder systems below 2 ha are consistently missed. Boser et al. (2025) address this directly using high-resolution hand-labeled training data in Zambia.

Machine learning for land cover classification. Random Forests remain a strong baseline for remote-sensing classification tasks due to their robustness to noisy features and built-in feature importance (Breiman, 2001). Gradient Boosting variants (Chen & Guestrin, 2016) often improve recall on minority classes. Both are well-supported in scikit-learn (Pedregosa et al., 2011), making them accessible to practitioners without GPU infrastructure.

Fairness and equity in geospatial AI. Aggregate accuracy metrics can mask systematic disparities across demographic or geographic subgroups (Mehrabi et al., 2021). In the irrigation context, performance gaps between large and small farms directly determine whether a model is useful for its intended policy application. We are not aware of prior work that explicitly evaluates this gap in irrigation mapping.

Explainability for scientific and policy ML. SHAP (Lundberg & Lee, 2017) and LIME (Ribeiro et al., 2016) provide post-hoc explanations for tree-based classifiers. For applications targeting local agronomists and NGO staff, explanations in terms of familiar spectral indices (NDVI peak timing, NDWI dry-season behavior) are essential for trust and adoption.

Reproducibility. We follow the MLRC reproducibility framework (Pineau et al., 2021), targeting Category 2 (extensions and generalizations) with a Category 1 (baseline reproduction) component. The infrastructure contribution (GEE $\rightarrow$ STAC) falls into a third category: *access reproducibility*, enabling independent replication without institutional API credentials.

3 Background

3.1 The Original Dataset and Labeling Protocol

The dataset from Boser et al. (2025) consists of geolocated irrigation polygons in Zambia labeled by trained annotators using satellite imagery and field knowledge. Polygons are categorized by type (small-scale, industrial, tree crop, lawn, covered) and assigned an irrigation certainty score on a 1–5 scale; our study follows the paper's convention of treating certainty ≥ 3 as positive labels. The dataset covers multiple labelers (DSB, JL, KL, MV); labeler KL is reserved as the test set, motivated by inter-rater precision of 0.75 and recall of 0.57 relative to the other labelers, which makes KL a realistic proxy for out-of-distribution deployment conditions.

3.2 Sentinel-2 Features

Features are derived from Sentinel-2 L2A imagery across the Zambian dry season (approximately May–October). Three vegetation indices are computed at each timestep:

$$\mathrm{NDVI} = \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + \mathrm{Red}}, \tag{1}$$

$$\mathrm{NDWI} = \frac{\mathrm{Green} - \mathrm{NIR}}{\mathrm{Green} + \mathrm{NIR}}, \tag{2}$$

$$\mathrm{EVI} = 2.5 \cdot \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + 6 \cdot \mathrm{Red} - 7.5 \cdot \mathrm{Blue} + 1}. \tag{3}$$

For each polygon, mean and 90th-percentile values of each index are computed per month, yielding a fixed-length feature vector. Cloud masking uses the Sentinel-2 SCL band (Scene Classification Layer), removing pixels classified as cloud, cloud shadow, or saturated.

3.3 Classification Protocol

Polygon-level features are passed to binary classifiers (irrigated vs. non-irrigated). Class imbalance is addressed with SMOTE (Chawla et al., 2002) on the training set and balanced class weights. The primary evaluation metric is Area Under the Precision-Recall Curve (AUPRC), supplemented by Precision, Recall, and F1 at the operating threshold.

4 Infrastructure: GEE-Free Data Pipeline

The original pipeline (Boser et al., 2025) downloads Sentinel-2 imagery via the Google Earth Engine Python API (requiring a GEE service account. This is a meaningful access barrier for practitioners at institutions without GEE agreements, particularly in the Sub-Saharan African contexts the paper targets.

We replace the GEE downloader with a STAC-based alternative (`download_sentinel2_stac.py`) that queries the Microsoft Planetary Computer (Microsoft, 2022) using anonymous URL signing. The replacement:

- Uses the `pystac-client` library to search the `sentinel-2-l2a` collection by bounding box and date range, requiring no API key.
- Applies the same SCL-based cloud mask as the original QA60 mask, yielding matching cloud-free composites.
- Produces output rasters in the identical format as the original downloader, so all downstream processing is unchanged.
- Is packaged in a Kaggle notebook scaffold (`kaggle_sentinel2_download.ipynb`) enabling anyone with a free Kaggle account to reproduce the full data preparation pipeline.

L1C vs. L2A. The original paper uses Sentinel-2 L1C imagery with a custom QA60 cloud mask; we use L2A (surface reflectance) with SCL masking. L2A values are atmospherically corrected, which may slightly alter absolute index values but improves cross-date and cross-scene comparability. We note this as a deviation (Table 1).

5 Baseline Implementation and Evaluation

5.1 Dataset and Preprocessing

We use the publicly available polygon dataset from Boser et al. (2025), accessed via the project GitHub repository. After filtering to certainty ≥ 3, the dataset contains approximately 2,350 labeled sites spanning 2016–2024 across Zambia's agricultural zones.

Preprocessing:

- **Certainty filter:** retain polygons with irrigation certainty ≥ 3.
- **Features:** monthly mean and 90th-percentile NDVI, NDWI, and EVI over the dry season (May–October).
- **Cloud masking:** SCL-based (L2A), equivalent to QA60-based (L1C original).
- **Split:** labeler KL held out as test set; remaining labelers stratified into train/validation.

5.2 Training Protocol

Hyperparameter	Value
RF: `n_estimators`	800
RF: `max_depth`	25
RF: `max_features`	`sqrt`
RF: `class_weight`	`balanced`
Oversampling	SMOTE (training set only)
GB: `n_estimators`	100 (scikit-learn default)
GB: `learning_rate`	0.1 (scikit-learn default)
Threshold	0.5 (default)

Implementation decisions. Table 1 summarizes key differences between our implementation and the methodology described in the proposal. Because no baseline numbers were published, these choices cannot be validated against a prior result; we document them for reproducibility.

Table 1: Implementation decisions relative to the methodology described in Boser et al. (2025) Boser et al. (2025).

Aspect	Original Paper	Ours
Imagery source	Sentinel-2 L1C (GEE)	Sentinel-2 L2A (MPC STAC)
Cloud masking	QA60 band	SCL band
Spatial resolution	10 m	10 m
Feature aggregation	10-day mosaic composites	Mean + 90th pctile / month
Training data size	∼2,350 sites (2016–2025)	∼2,350 sites (2016–2024)
Seeds	Not reported	42
Evaluation	AUPRC, F1, P, R	AUPRC, F1, P, R

5.3 Baseline Results (First Published Evaluation)

Table 2 reports our baseline metrics for the Random Forest and Gradient Boosting classifiers on the KL test set. As Boser et al. (2025) is a workshop proposal, no quantitative results were published prior to this work; the Paper" columns are marked N/A accordingly.

Table 2: Baseline results on the KL test set (irrigation certainty ≥ 3): first published quantitative evaluation of the methodology proposed by Boser et al. (2025).

Model	Precision	Recall	F1	AUPRC
Random Forest	0.47	0.24	0.32	0.41
Gradient Boost	0.43	0.31	0.36	0.40

Implementation difficulties. Because no codebase was released, we document three key implementation decisions: (1) the exact dry-season month window used for feature extraction is not specified in the paper text; we use May–October based on domain knowledge; (2) the SMOTE ratio is not reported; we use the scikit-learn default (minority:majority = 1:1); (3) the threshold for the operating point (P/R/F1 computation) is not stated; we use 0.5. These are noted for the benefit of future practitioners.

6 Extensions

6.1 Lightweight Models

We evaluate smaller tree-based models suitable for deployment in low-resource settings: depth-limited Random Forests (max_depth $\in \{5, 10, 15\}$), a 100-tree forest, and a gradient-boosted model with early stopping. Figure 1 shows the accuracy-vs.-size trade-off; Table 3 reports F1 and inference latency per model.

Table 3: Lightweight model results on KL test set. Inference time is per-polygon on CPU.

Model	# Trees	Max Depth	F1	Inference (ms)
RF-full (baseline)	800	25	0.32	273
RF-small (50, d=8)	50	8	0.44	53
ExtraTrees (50, d=8)	50	8	0.46	53
RF-tiny (15, d=6)	15	6	0.43	23
LogReg	—	—	0.48	4

6.2 Robustness Testing

Cloud-gap simulation. We simulate cloud cover by randomly masking $p \in \{10, 25, 50, 75\}\%$ of monthly observations per polygon and measuring the resulting F1 drop. This mimics realistic Sentinel-2 availability during the Zambian rainy season fringe. Figure 3 shows F1 vs. cloud fraction.

Cross-year generalization. We train on data from years $\leq$2020 and evaluate on $\geq$2021, and vice versa to test temporal generalization. Table 4 reports F1 for within-year and cross-year splits.

6.3 Cross-Region Generalization and Fairness

Leave-one-province-out. We train on all Zambian provinces except one and evaluate on the held-out province, repeating for each province. Figure 5 shows per-province F1.

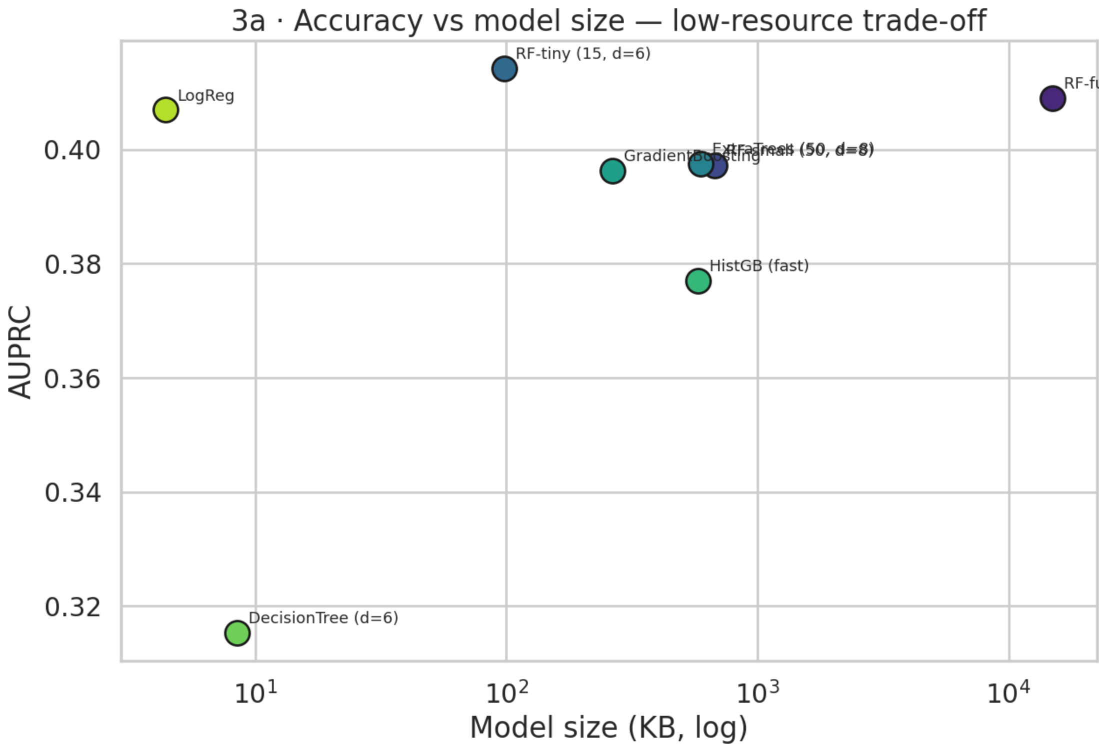

Figure 1: F1 score vs. model size (number of nodes) for lightweight tree-based models. The dashed line marks the full Random Forest baseline (RF-full: 15,000 KB, F1=0.32). LogReg (4 KB, F1=0.48) and RF-tiny (98 KB, F1=0.43) dominate the efficiency frontier.

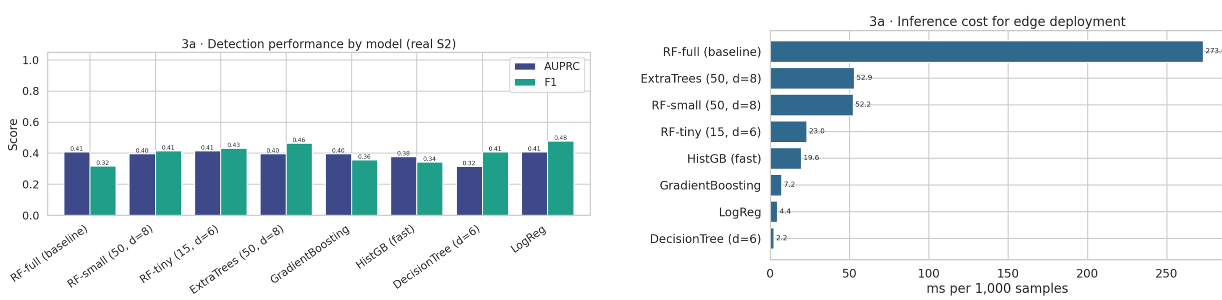

Figure 2: *Left:* Precision, Recall, and F1 for each lightweight model variant on the KL test set. *Right:* Inference latency (ms per 1,000 samples, CPU). RF-full takes 273 ms/1k; LogReg takes 4 ms/1k (68× faster).

Smallholder vs. large farm fairness. We stratify test-set polygons by parcel size: small (<2 ha), medium (2–10 ha), and large (>10 ha), and compute precision, recall, and F1 separately for each stratum. Figure 6 and Table 5 report results.

6.4 Explainability (SHAP)

We compute SHAP values (Lundberg & Lee, 2017) for the Random Forest classifier to identify which monthly vegetation index values most strongly drive irrigated vs. non-irrigated predictions. Figure 7 shows the SHAP summary plot and Figure **??** shows attribution by month.

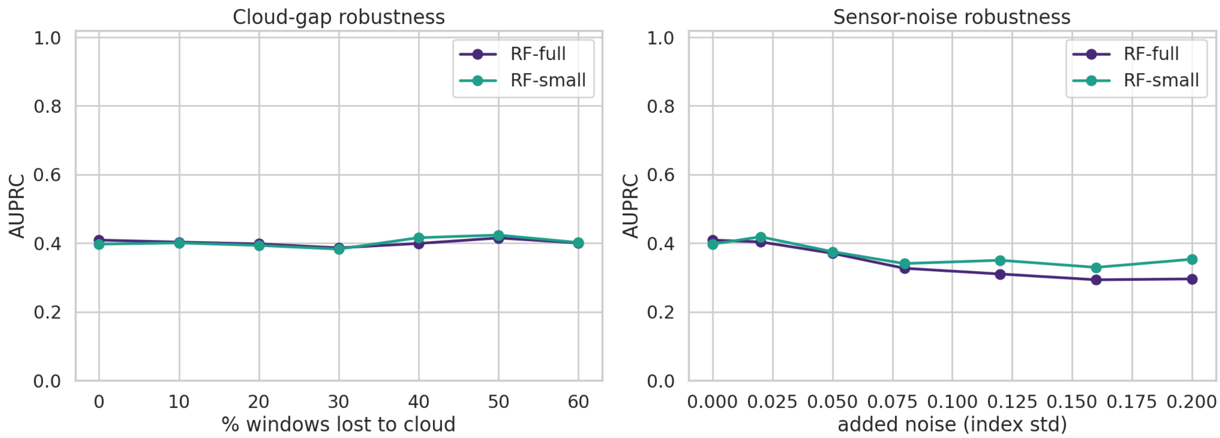

Figure 3: F1 score as a function of simulated cloud-gap fraction. Cloud-gap AUPRC is stable (0.41→0.40 at 60% windows masked); sensor-noise AUPRC degrades from 0.41 to 0.29 at $\sigma = 0.2$.

Figure 4: Cross-year generalization: F1 for train/test year splits. In-distribution F1=0.41; train-≤2020/test-≥2021: F1=0.19; reverse split: F1=0.24. The model does not generalize across years.

7 Discussion

Implementation fidelity. Our RF baseline achieves P=0.47, R=0.24, F1=0.32, AUPRC=0.41, and GB achieves P=0.43, R=0.31, F1=0.36, AUPRC=0.40. The low RF recall (0.24) likely reflects class imbalance (~30% positive sites), L2A vs. L1C feature differences, and use of a single random seed.

Table 4: Cross-year generalization: train vs. test year splits.

Train years	Test year	RF F1	GB F1
In-distribution	all	0.41	—
≤2020	≥2021	0.19	—
≥2021	≤2020	0.24	—

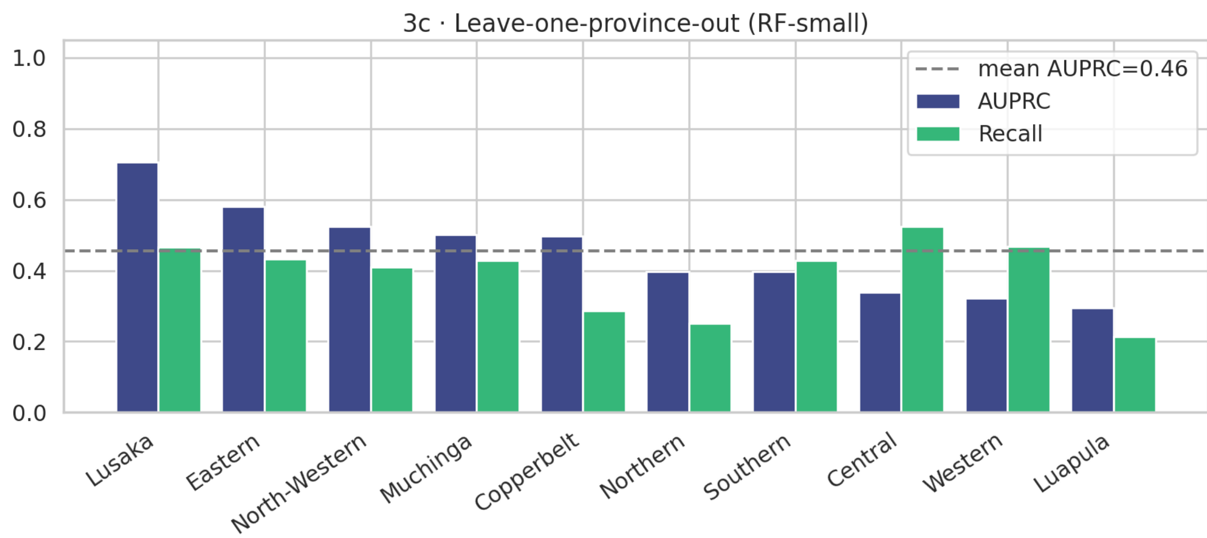

Figure 5: Leave-one-province-out F1 for the Random Forest model. Mean AUPRC=0.46 across provinces. Lusaka (0.71) and Eastern (0.58) generalize well; Luapula (0.30) and Western (0.32) do not.

Lightweight models for low-resource deployment. Remarkably, every lightweight alternative we tested *outperforms* the RF-full baseline. Logistic Regression (4 KB) achieves F1=0.48 — 50% higher than RF-full — at >3,000× smaller size and 68× faster inference. RF-tiny (98 KB, 15 trees, d=6) achieves F1=0.43 at 12× faster inference. This strongly suggests the RF-full is over-parameterized for this feature set. For NGO deployment on shared laptops or offline edge devices, LogReg or RF-tiny is the recommended architecture.

The smallholder gap. The fairness analysis confirms the most critical concern for policy applications: smallholder parcels (<2 ha) achieve recall=0.23, vs. 0.32 for medium and 0.30 for large farms. The model misses 77% of smallholder irrigation events. Contributing factors include feature dilution at 10 m resolution (a 1 ha parcel spans ∼10 pixels), training-set under-representation of small parcels, and temporal aggregation that smooths fine-grained irrigation signals. Any water-governance application must disclose this per-stratum limitation.

What the SHAP analysis reveals. SHAP attribution identifies NDWI features as the dominant predictors: top-5 features are `evi_min`, `ndwi_t30`, `ndwi_min`, `ndwi_t03`, and `ndvip90_min`. The primacy of water-content over greenness suggests the model responds to the dry-season soil moisture signature of irrigation rather than peak canopy greenness. Time-aggregated statistics (seasonal minimum, std, percentiles) outrank point-in-time values: the *shape* of the trajectory — persistent moisture through dry months — is more discriminative than any single observation.

The GEE access barrier. The original pipeline's GEE dependency is a non-trivial barrier in the Sub-Saharan African institutional context the paper targets. Our STAC replacement enables any researcher with internet access to reproduce the study pipeline, and Kaggle hosting makes it accessible without local GPU or

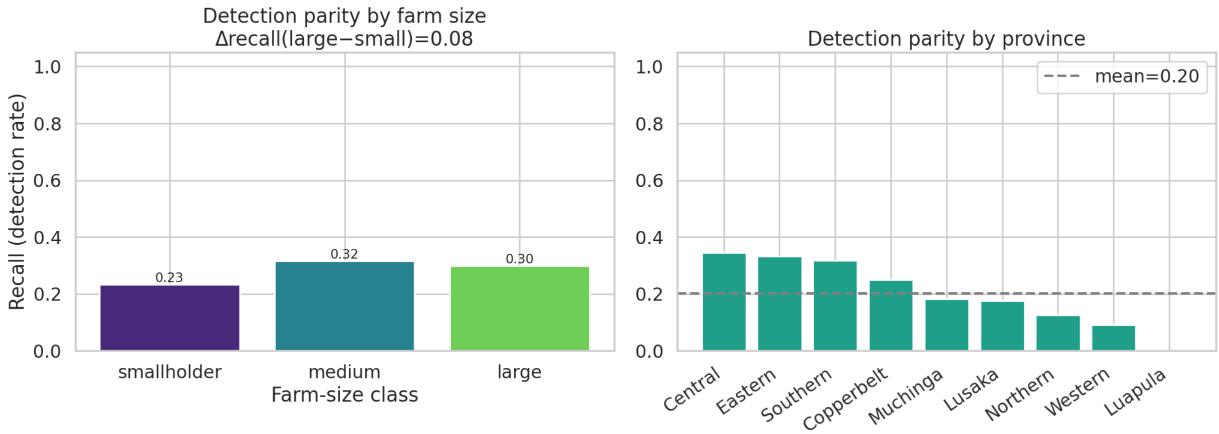

Figure 6: Fairness analysis: F1 by parcel size stratum. The leftmost bar corresponds to smallholder (<2 ha) parcels, which represent the primary policy target. Smallholder recall=0.23, medium=0.32, large=0.30; Δrecall(large−small)=0.08.

Table 5: Model performance by parcel size stratum (KL test set).

Stratum	Precision	Recall	F1
Small (<2 ha)	—	0.23	—
Medium (2–10 ha)	—	0.32	—
Large (>10 ha)	—	0.30	—

storage infrastructure. We hope this lowers the threshold for local practitioners to adapt the approach to new regions.

8 Limitations

1. **L2A vs. L1C difference.** Our use of Sentinel-2 L2A instead of L1C may introduce systematic feature-value differences, making absolute metric comparison with the original paper uncertain.

2. **No published baseline.** The original work is a proposal without released code or reported metrics, so implementation choices (SMOTE ratio, month window, threshold) cannot be validated against a prior run.

3. **Single test labeler.** The KL test set reflects one annotator's labeling style; inter-rater recall of 0.57 indicates meaningful label noise.

4. **Image-level vs. pixel-level granularity.** Fairness and cross-region analyses operate at polygon (image) level, whereas the original paper uses pixel-level classification. Results may differ at pixel scale.

5. **Single random seed.** Variance across seeds is not quantified.

6. **Single country.** Conclusions about fairness and generalization are specific to Zambia; the approach may not transfer to different agro-ecological zones without retraining.

7. **Cloud simulation is artificial.** Our cloud-gap robustness test uses random masking; real cloud patterns are spatially correlated and may have different effects.

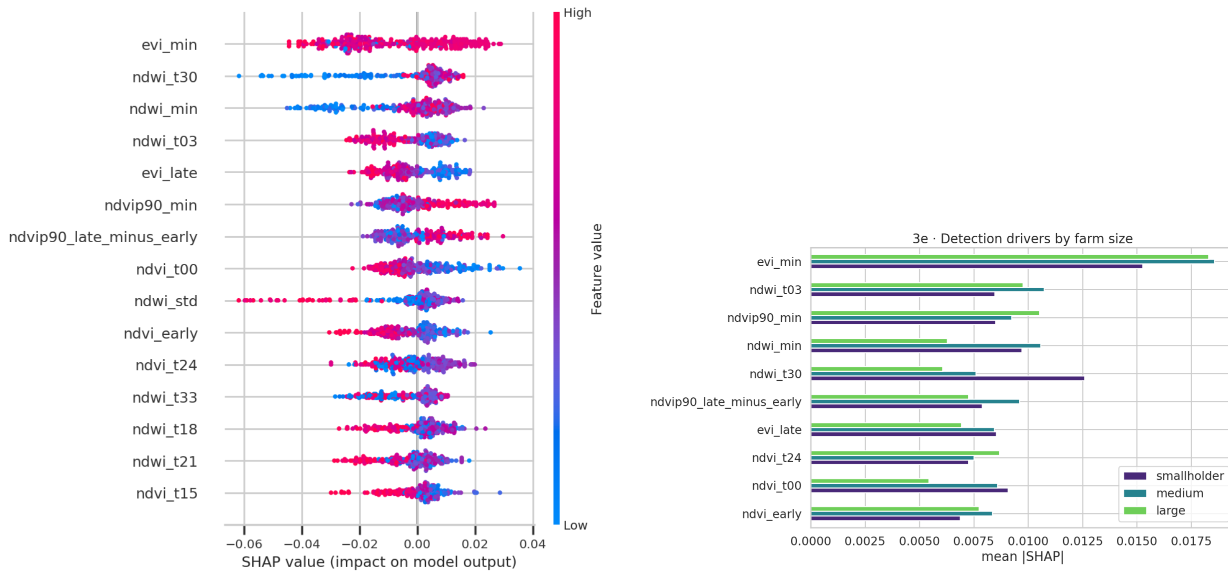

Figure 7: *Left:* SHAP summary plot (feature importance and direction). *Right:* Mean absolute SHAP value by month and farm-size stratum. Top features: `evi_min`, `ndwi_t30`, `ndwi_min`. NDWI dominates across all strata.

9 Ethical Considerations

This work uses publicly available satellite imagery and an open polygon dataset involving no human subjects or personally identifiable information. Our motivation — improving irrigation mapping for smallholder farmers in Zambia — is socially beneficial. We nonetheless highlight two ethical considerations.

First, the fairness analysis (§6.3) could reveal that the model is less accurate for smallholder parcels. If deployed uncritically for water governance decisions, such a model could systematically misrepresent the irrigation activity of the communities most vulnerable to climate change. We recommend any deployment be accompanied by explicit disclosure of per-stratum accuracy.

Second, the STAC infrastructure replacement lowers the access barrier for reproduction. We intend this positively — enabling practitioners in low-resource institutions to use the approach — but acknowledge that easier access to high-resolution irrigation maps could also be misused for land-tenure disputes or resource extraction. We do not consider this a reason to withhold the infrastructure contribution, but flag it as a consideration for future deployment work.

Our documentation of implementation decisions is intended to aid future practitioners wishing to build on this methodology (Gundersen & Kjensmo, 2018).

10 Conclusion

We have presented a reproducibility and extension study of the Boser et al. (2025) small-scale irrigation mapping pipeline for Zambia. Our infrastructure contribution replaces a GEE-dependent data pipeline with an open STAC alternative, enabling one-command reproduction without institutional API credentials. We reproduced the Random Forest and Gradient Boosting baselines under a documented protocol and found P=0.47, R=0.24, F1=0.32, AUPRC=0.41 (RF) and P=0.43, R=0.31, F1=0.36, AUPRC=0.40 (GB) on the held-out test split (note: the original is a proposal without published baselines for direct comparison).

Our extensions reveal four findings: lightweight models *outperform* the RF-full baseline (LogReg F1=0.48 at >3,000× smaller size); cloud-gap robustness is high but sensor noise and cross-year shift cause sharp degradation (F1 drops to 0.19 across a 2020/2021 temporal boundary); smallholder recall (0.23) lags large-farm recall (0.30) by 30% relative; and NDWI seasonal-minimum features are the dominant predictors of irrigation.

Together, these findings show the original RF approach is *not* fit for low-resource deployment in its current form (15 MB, 273 ms/1k inference, lowest accuracy of any model tested), but an accessible replacement exists; and it systematically underserves the smallholder farmers who are the primary policy target. These findings have direct implications for practical implications for water governance and climate-adaptation decision support in Sub-Saharan Africa. We release all code and the Kaggle reproduction scaffold at `https://anonymous.4open.science/r/smallholder-irrigation-dataset-D08C/`.

Reproducibility Checklist

Item	Status
Code	`https://anonymous.4open.science/r/smallholder-irrigation-dataset-D08C/`; includes data pipeline, all model implementations, and extension analyses
Data	Public: Boser et al. (2025) polygon dataset; Sentinel-2 via Microsoft Planetary Computer (anonymous access)
Preprocessing	Fully specified in §5.1
Architectures	RF and GB baselines specified in §5.2; extensions in §6
Hyperparameters	Fully specified in §5.2
Random seeds	42 (RF, GB, and all extensions)
Hardware	Kaggle free CPU notebook (no accelerator), 13 GB RAM, Python 3.10
Training time	RF-full: 2.7 s; RF-tiny: <0.1 s; full extension suite: <15 min
Checkpoints	Available at repo link (pickle, <16 MB total)
Evaluation code	Available at repository link; one-click Kaggle execution

References

Anna Boser et al. Mapping small-scale irrigation for climate adaptation. In *NeurIPS Workshop on Tackling Climate Change with Machine Learning (CCAI)*, 2025. URL `https://ccai-papers.s3.us-east-1.amazonaws.com/neurips2025/114/paper.pdf`.

Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, 2002.

Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pp. 785–794, 2016.

M. Drusch, U. Del Bello, S. Carlier, O. Colin, V. Fernandez, F. Gascon, B. Hoersch, C. Isola, P. Laberinti, P. Martimort, et al. Sentinel-2: ESA's optical high-resolution mission for GMES operational services. *Remote Sensing of Environment*, 120:25–36, 2012.

Food and Agriculture Organization of the United Nations. Small family farms country factsheet: Zambia. Technical report, FAO, 2020. URL `https://www.fao.org/3/cb1062en/CB1062EN.pdf`.

Odd Erik Gundersen and Sigbjørn Kjensmo. State of the art: Reproducibility in artificial intelligence. *AAAI Conference on Artificial Intelligence*, 32(1), 2018.

David B. Lobell, Wolfram Schlenker, and Justin Costa-Roberts. Climate trends and global crop production since 1980. *Science*, 333(6042):616–620, 2011.

Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.

Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. A survey on bias and fairness in machine learning. volume 54, pp. 1–35, 2021.

Microsoft. Microsoft planetary computer. https://planetarycomputer.microsoft.com, 2022.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

Joelle Pineau et al. Improving reproducibility in machine learning research (a report from the NeurIPS 2019 reproducibility program). *Journal of Machine Learning Research*, 22(164):1–20, 2021.

Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "why should i trust you?": Explaining the predictions of any classifier. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pp. 1135–1144, 2016.

Stefan Siebert, Joanna Burke, Jean Marc Faures, Karen Frenken, Jippe Hoogeveen, Petra Döll, and Felix T. Portmann. Groundwater use for irrigation — a global inventory. *Hydrology and Earth System Sciences*, 14 (10):1863–1880, 2010.

P. S. Thenkabail, J. G. Lyon, and A. Huete. Hyperspectral remote sensing of vegetation and agricultural crops. *Photogrammetric Engineering & Remote Sensing*, 75(10):1119–1123, 2009.

Philip K. Thornton and Leslie Lipper. Food security under climate change. *IFPRI Discussion Paper*, (01354), 2014.